

Charm TUI Ecosystem & Python Alternatives - Comprehensive Research for Hekat DSL CLI

Research Date: 2025-10-20 **Focus:** TUI/CLI design patterns for Hekat Phase 5-6 development **Target:** Interactive DSL builder, syntax highlighting, real-time validation, execution dashboard ---

Executive Summary

Key Findings

1. **Charm Ecosystem (Go)** represents the gold standard for modern TUI applications with its elegant Elm-inspired MVU architecture, comprehensive component library, and production-proven applications (GitHub CLI, Glow, VHS) 2. **Textual (Python)** is the best Python alternative, offering 95% of Charm's capabilities with a web-inspired API, comprehensive widget library, CSS-based styling, and async-first architecture 3. **Rich (Python)** provides beautiful terminal output but lacks full TUI capabilities - best used as a foundation library (Textual is built on Rich) 4. **Prompt Toolkit (Python)** excels at interactive command-line interfaces but requires more manual work for full TUI applications

Recommendations for Hekat

Primary Recommendation: Build Hekat TUI using **Textual** with **Rich** for enhanced output formatting **Rationale:** - Native Python integration (Hekat's implementation language) - Mature, actively maintained (2024+) - Comprehensive widget library covers all Hekat needs - Web-inspired API familiar to developers - Excellent documentation and real-world examples - Async-

first design matches Hekat's architecture - Can run in terminal AND browser - 16.7M colors, mouse support, smooth animations **Implementation Approach:** - **Phase 5:** Interactive DSL builder wizard using Textual forms - **Phase 6:** Real-time execution dashboard with live logs and progress ---

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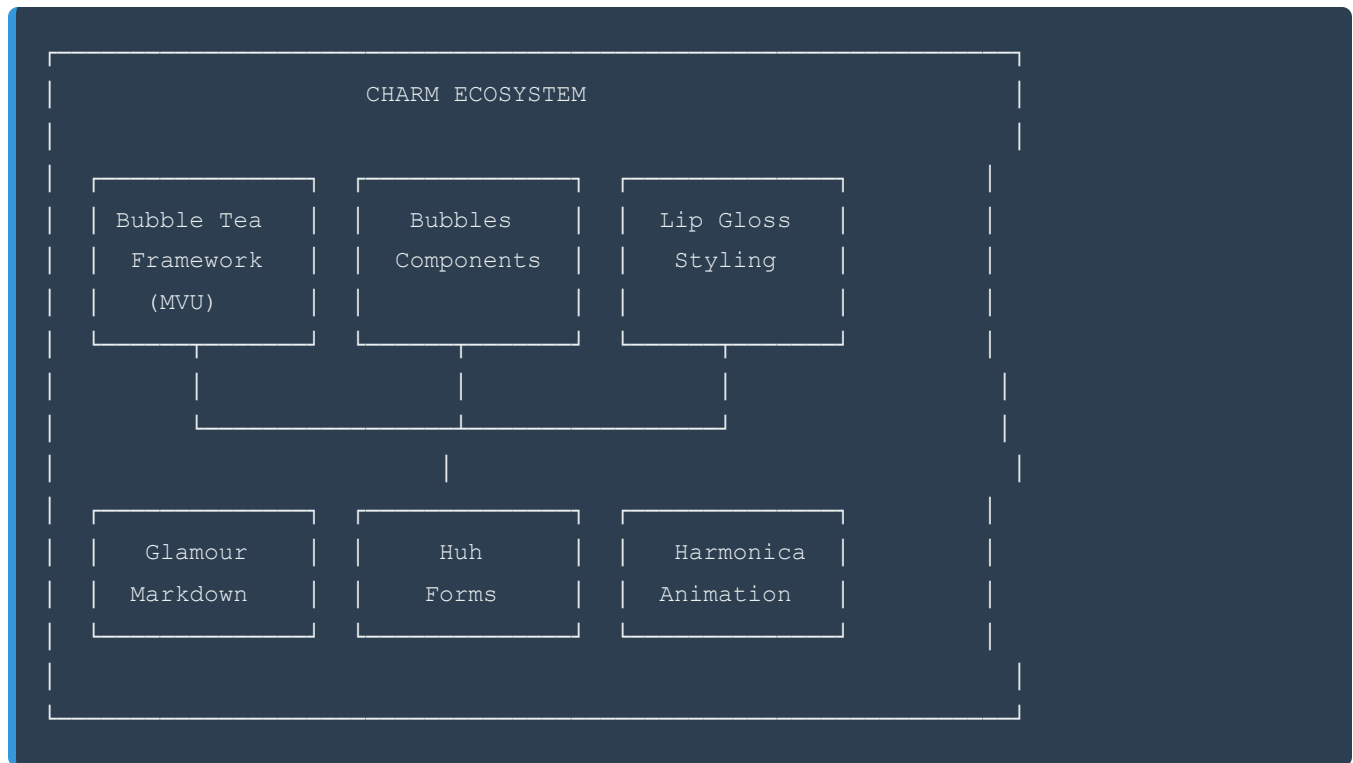
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Charm Ecosystem Overview

What is Charm?

Charm creates elegant, production-ready terminal applications that make "the command line glamorous." Their ecosystem includes everything needed to build sophisticated TUI applications.

Core Libraries



1. Bubble Tea - Core TUI Framework **Purpose:** Foundation for building interactive terminal applications **Architecture:** Elm MVU (Model-View-Update) **Version:** 1.0+ (2024)

Used By: Min.io, Supabase, GitHub CLI **Key Features:** - Model-View-Update architecture - Event-driven programming - Keyboard and mouse support - Async command execution - Component composition - Terminal control abstraction

2. Bubbles - Component Library **Purpose:** Reusable TUI components for Bubble Tea **Components:** - `input` - Text input fields - `textarea` - Multi-line text editing - `list` - Scrollable lists with filtering - `table` - Data tables with sorting - `viewport` - Scrollable content areas - `spinner` - Loading indicators - `progress` - Progress bars - `paginator` - Page navigation - `filepicker` - File selection - `timer` - Countdown/stopwatch - `help` - Keyboard help display - `key` - Keyboard input handling

3. Lip Gloss - Styling & Layout **Purpose:** CSS-like styling for terminal output **Inspiration:** Web CSS but adapted for terminal constraints **Features:** - Color support (256-color, 24-bit RGB) - Border styles (rounded, thick, double, etc.) - Padding and margins - Text alignment - Width and height constraints - Foreground/background colors - Text decorations (bold, italic, underline) - Horizontal and vertical joining - Table rendering **Example Style:**

```
var style = lipgloss.NewStyle().
    Bold(true).
    Foreground(lipgloss.Color("#FAFAFA")).
    Background(lipgloss.Color("#7D56F4")).
    PaddingTop(2).
    PaddingLeft(4).
    Width(22)
```

4. Glamour - Markdown Rendering **Purpose:** Beautiful markdown rendering in terminals
Used By: GitHub CLI (gh), Glow **Features:** - Multiple built-in themes (dark, light, auto-detect) - Custom style sheets - Syntax highlighting for code blocks - Responsive word wrapping - Link rendering - Table support - Image placeholders **Use Cases for Hekat:** - DSL documentation viewer - Agent/command help system - Tutorial and examples browser - Release notes and changelogs

5. Huh - Interactive Forms **Purpose:** Simple, powerful forms and wizards
Latest: Modern, production-ready (2024) **Form Components:** - Text input - Text area - Select (single choice) - Multi-select - Confirm (yes/no) - File picker - Note (informational text)
Features: - Multi-step wizards - Field validation - Conditional fields - Accessible design - Keyboard navigation - Error display - Theme support **Perfect for Hekat DSL Builder ---**

Architecture Deep Dive: Elm MVU Pattern

What is MVU (Model-View-Update)?

The MVU pattern, originating from Elm functional programming language, provides a clean separation of concerns for interactive applications.

Core Concepts



Three Core Functions

1. Init() - Initialization **Purpose**: Set up initial state and commands

```
func (m model) Init() tea.Cmd {
    return nil // or initial command
}
```

Returns: Command to execute on startup (or nil) ##### 2. Update(msg tea.Msg) - State Transitions **Purpose:** Handle all events and update state

```
func (m model) Update(msg tea.Msg) (tea.Model, tea.Cmd) {
    switch msg := msg.(type) {
    case tea.KeyMsg:
        switch msg.String() {
        case "q", "ctrl+c":
            return m, tea.Quit
        case "up", "k":
            m.cursor--
        case "down", "j":
            m.cursor++
        }
    case tickMsg:
        m.counter++
        return m, tick()
    }
    return m, nil
}
```

Parameters: Message (event) **Returns:** Updated model + command to execute ##### 3. View() - Rendering **Purpose:** Render UI based on current state

```
func (m model) View() string {
    s := "Counter: " + strconv.Itoa(m.counter) + "\n\n"
    s += "Press q to quit.\n"
    return s
}
```

Parameters: Current model **Returns:** String representation of UI

Message Types

Messages are the events that flow through the system:

```
// Keyboard events
type tea.KeyMsg

// Mouse events
type tea.MouseMsg

// Window resize
type tea.WindowSizeMsg

// Custom messages
type tickMsg time.Time
type dataLoadedMsg []Item
type errorMsg error
```

Commands (Side Effects)

Commands represent async operations:

```
// Command signature
type Cmd func() Msg

// Example: API call
func fetchData() tea.Msg {
    data, err := api.GetData()
    if err != nil {
        return errorMsg{err}
    }
    return dataLoadedMsg{data}
}
```

Benefits of MVU

1. **Predictability:** Pure functions, no hidden state
2. **Testability:** Easy to unit test each function
3. **Debugging:** Clear event flow, easy to trace
4. **Composition:** Components can be nested
5. **Time Travel:** Can replay events for debugging
6. **Concurrent:** Safe message passing

MVU vs Traditional Imperative

Imperative (Traditional): - Mutable state scattered across code - Side effects everywhere - Event handlers modify state directly - Hard to test and debug
MVU (Functional): - Immutable state in one place - Side effects isolated in commands - Pure functions for transformations - Easy to test and reason about ---

Charm Libraries Detailed Analysis

Bubble Tea Framework

Key Capabilities
1. Terminal Control - Full screen or inline rendering - Alternate screen buffer - Mouse capture (click, drag, scroll) - Keyboard enhancement protocol - Focus/blur events - Bracketed paste support
2. Event System

```
// Keyboard events
tea.KeyMsg{
    Type: tea.KeyRunes, // Regular character
    Runes: []rune{'a'},
    Alt: false,
}

// Mouse events
tea.MouseMsg{
    X: 10, Y: 5,
    Button: tea.MouseButtonLeft,
    Action: tea.MouseActionPress,
}

// Window events
tea.WindowSizeMsg{
    Width: 80,
    Height: 24,
}
```

3. Component Composition Components are just models that implement Init/Update/View:

```

type model struct {
    input    textinput.Model
    list     list.Model
    viewport viewport.Model
}

func (m model) Update(msg tea.Msg) (tea.Model, tea.Cmd) {
    var cmds []tea.Cmd

    // Update child components
    m.input, cmd = m.input.Update(msg)
    cmds = append(cmds, cmd)

    m.list, cmd = m.list.Update(msg)
    cmds = append(cmds, cmd)

    return m, tea.Batch(cmds...)
}

```

4. Async Operations

```

// Long-running operation
func loadDataCmd() tea.Cmd {
    return func() tea.Msg {
        data := fetchFromAPI() // Could take seconds
        return dataLoadedMsg{data}
    }
}

// Timer/tick
func tick() tea.Cmd {
    return tea.Tick(time.Second, func(t time.Time) tea.Msg {
        return tickMsg(t)
    })
}

```

Advanced Features (v2.0+) **Enhanced Keyboard Support:** - Key press and key release events - Shift+Enter, Ctrl+Shift+A, etc. - Uniform keyboard layout handling - Kitty keyboard protocol **Mouse Enhancements:** - Cell motion (drag events) - All motion (hover events) -

Wheel scrolling - Button distinction **Focus Management:** - Terminal focus/blur detection - Bracketed paste events - Multiple window support

Bubbles Components Deep Dive

Input Component **Use Cases:** Single-line text input, search fields, form fields

```
input := textinput.New()
input.Placeholder = "Enter agent name..."
input.Focus()
input.CharLimit = 64
input.Width = 30
input.Prompt = "> "
input.PromptStyle = lipgloss.NewStyle().Foreground(lipgloss.Color("63"))
```

Features: - Placeholder text - Character limit - Cursor position control -

Suggestions/autocomplete - Validation - Echo mode (password masking) #### TextArea Component **Use Cases:** Multi-line editing, code input, DSL editing

```
textarea := textarea.New()
textarea.Placeholder = "# Enter Hekat DSL here..."
textarea.SetWidth(80)
textarea.SetHeight(20)
textarea.ShowLineNumbers = true
textarea.KeyMap = textarea.KeyMap{...} // Custom keys
```

Features: - Line numbers - Soft/hard wrapping - Cursor navigation - Selection support -

Undo/redo - Custom key bindings #### List Component **Use Cases:** Menu selection, file browser, agent picker

```

type item struct {
    title, desc string
}

func (i item) Title() string      { return i.title }
func (i item) Description() string { return i.desc }
func (i item) FilterValue() string { return i.title }

list := list.New(items, list.NewDefaultDelegate(), 20, 10)
list.Title = "Select Agent"
list.SetShowStatusBar(true)
list.SetFilteringEnabled(true)

```

Features: - Filtering/search - Pagination - Status bar - Custom item rendering - Selection callbacks - Keyboard navigation (j/k, arrows) ##### Table Component **Use Cases:** Data display, logs, execution history

```

columns := []table.Column{
    {Title: "Agent", Width: 20},
    {Title: "Status", Width: 10},
    {Title: "Duration", Width: 10},
}

rows := []table.Row{
    {"agent-1", "✓ Done", "2.3s"},
    {"agent-2", "⚙ Running", "1.1s"},
}

t := table.New(
    table.WithColumns(columns),
    table.WithRows(rows),
    table.WithFocused(true),
    table.WithHeight(10),
)

```

Features: - Sortable columns - Fixed/scrollable rows - Row selection - Custom cell rendering - Header styling - Focus management ##### Viewport Component **Use Cases:** Log viewer, documentation, scrollable content

```
vp := viewport.New(80, 20)
vp.SetContent(longContent)
vp.YOffset = 0 // Scroll position
vp.MouseWheelEnabled = true
```

Features: - Smooth scrolling - Mouse wheel support - Keyboard navigation (PgUp/PgDn) - Dynamic content updates - Scroll indicators ##### Progress & Spinner Components **Use Cases:** Task progress, loading states

```
// Progress bar
progress := progress.New(progress.WithDefaultGradient())
progress.Width = 40
progress.ShowPercentage = true

// Spinner
spinner := spinner.New()
spinner.Spinner = spinner.Dot
spinner.Style = lipgloss.NewStyle().Foreground(lipgloss.Color("205"))
```

Lip Gloss Styling System

Style Building

```
// Define a style
titleStyle := lipgloss.NewStyle().
    Bold(true).
    Foreground(lipgloss.Color("#FFFFFF")).
    Background(lipgloss.Color("#7D56F4")).
    Padding(0, 1).
    MarginTop(1).
    Width(50).
    Align(lipgloss.Center)

// Apply style
title := titleStyle.Render("Hekat DSL Builder")
```

Layout Utilities **Horizontal Join:**

```

leftPanel := lipgloss.NewStyle().
    Width(30).
    Border(lipgloss.RoundedBorder()).
    Render("Left content")

rightPanel := lipgloss.NewStyle().
    Width(50).
    Border(lipgloss.RoundedBorder()).
    Render("Right content")

layout := lipgloss.JoinHorizontal(
    lipgloss.Top,
    leftPanel,
    rightPanel,
)

```

Vertical Join:

```

header := headerStyle.Render("Header")
body := bodyStyle.Render("Body")
footer := footerStyle.Render("Footer")

layout := lipgloss.JoinVertical(
    lipgloss.Left,
    header,
    body,
    footer,
)

```

Border Styles

```

border := lipgloss.RoundedBorder() // ┌─┐
border := lipgloss.ThickBorder()   // ┌─┐
border := lipgloss.DoubleBorder()  // ┌─┐
border := lipgloss.NormalBorder()  // ┌─┐

```

Color Support

```
// 256-color palette
style.Foreground(lipgloss.Color("205"))

// RGB colors
style.Foreground(lipgloss.Color("#FF5733"))

// Adaptive colors (auto light/dark)
style.Foreground(lipgloss.AdaptiveColor{
    Light: "#000000",
    Dark:  "#FFFFFF",
})
```

Huh Forms Library

Form Building

```

form := huh.NewForm(
    huh.NewGroup(
        // Text input
        huh.NewInput().
            Key("name").
            Title("Agent Name").
            Placeholder("my-agent").
            Validate(validateName),

        // Select (dropdown)
        huh.NewSelect[string]().
            Key("type").
            Title("Agent Type").
            Options(
                huh.NewOption("Python", "python"),
                huh.NewOption("Go", "go"),
                huh.NewOption("Node.js", "nodejs"),
            ),

        // Multi-select (checkboxes)
        huh.NewMultiSelect[string]().
            Key("features").
            Title("Features").
            Options(
                huh.NewOption("Async", "async"),
                huh.NewOption("Parallel", "parallel"),
                huh.NewOption("Streaming", "streaming"),
            ),

        // Confirm (yes/no)
        huh.NewConfirm().
            Key("confirm").
            Title("Create agent?").
            Affirmative("Yes").
            Negative("No"),
    ),
)

// Run form
err := form.Run()

// Get values
name := form.GetString("name")

```

```
agentType := form.GetString("type")
features := form.Get("features").([]string)
```

Multi-Step Wizards

```
form := huh.NewForm(
    // Step 1: Basic info
    huh.NewGroup(
        huh.NewInput().Key("name").Title("Name"),
        huh.NewInput().Key("desc").Title("Description"),
    ).Title("Basic Information"),

    // Step 2: Configuration
    huh.NewGroup(
        huh.NewSelect[string]().Key("runtime").Title("Runtime"),
        huh.NewInput().Key("timeout").Title("Timeout"),
    ).Title("Configuration"),

    // Step 3: Review
    huh.NewGroup(
        huh.NewNote().
            Title("Review").
            Description("Please review your configuration..."),
        huh.NewConfirm().Key("confirm").Title("Create?"),
    ).Title("Confirmation"),
)
```

Validation

```
func validateName(s string) error {
    if len(s) < 3 {
        return errors.New("name must be at least 3 characters")
    }
    if !regexp.MustCompile(`^[a-z-]+$`).MatchString(s) {
        return errors.New("name must contain only lowercase letters and hyphens")
    }
    return nil
}

huh.NewInput().
    Key("name").
    Title("Agent Name").
    Validate(validateName)
```

Glamour Markdown Renderer

Basic Usage

```
import "github.com/charmbracelet/glamour"

// Auto-detect terminal theme
r, _ := glamour.NewTermRenderer(
    glamour.WithAutoStyle(),
    glamour.WithWordWrap(80),
)

out, _ := r.Render(markdown)
fmt.Print(out)
```

Custom Themes

```
// Use built-in theme
r, _ := glamour.NewTermRenderer(
    glamour.WithStylePath("dark"), // or "light", "notty", "pink"
)

// Custom JSON style
r, _ := glamour.NewTermRenderer(
    glamour.WithStylesFromJSONFile("custom.json"),
)

```

Use Cases for Hekat 1. **Agent Documentation:**

```
// Display agent help
agentDoc := getAgentMarkdown(agentName)
r.Render(agentDoc)

```

2. DSL Reference:

```
// Show DSL syntax guide
dslGuide := loadMarkdown("docs/dsl-reference.md")
r.Render(dslGuide)

```

3. Interactive Help System:

```
// Help browser with search
type helpBrowser struct {
    topics    []string
    selected  int
    content   string
}

func (h helpBrowser) View() string {
    // List topics on left
    // Rendered markdown on right (Glamour)
}

```

Visual Design Patterns

Charm Applications UX Analysis

Glow - Markdown Reader **Layout Pattern:**



Design Elements: - Clean, minimal interface - Single focus area - Keyboard shortcuts always visible - Responsive to terminal size - Beautiful markdown rendering - Syntax highlighting for code blocks

GitHub CLI (gh) - Multi-Panel Layout **Layout Pattern:**

Pull Requests		●●● 1/3
Filters:	#123: Add new feature	
○ All	✓ CI passing • 2 reviews • Ready to merge	
● Open		
○ Closed	Description:	
	This PR adds a new feature that enables...	
Sort:		
○ Created	Files changed: 12	
● Updated	+234 -45	
	[View diff] [Merge] [Close]	
↑/↓ navigate • enter select • tab switch • ? help		

Design Elements: - Two-panel layout (filter + content) - Status indicators (✓, ●, ○) - Action buttons - Contextual help bar - Tab navigation between sections ##### VHS - Terminal Recorder **Layout Pattern:**

VHS - Recording	
Output: demo.gif	
Settings: Width: 800px, Height: 600px	
● Recording...	
	67%
Frame: 234/350	
Elapsed: 12.3s	

Design Elements: - Progress indicators - Real-time stats - Minimal distraction during recording - Status emoji (●, ✓, II)

Color Schemes

Dark Theme (Default)

```
var (  
    // Primary colors  
    Primary    = lipgloss.Color("#7D56F4") // Purple  
    Secondary  = lipgloss.Color("#FF6AC1") // Pink  
    Accent     = lipgloss.Color("#00D9FF") // Cyan  
  
    // Status colors  
    Success    = lipgloss.Color("#04B575") // Green  
    Warning    = lipgloss.Color("#FFAA00") // Orange  
    Error      = lipgloss.Color("#EE4B2B") // Red  
    Info       = lipgloss.Color("#00A9FF") // Blue  
  
    // Text colors  
    TextPrimary   = lipgloss.Color("#FFFFFF")  
    TextSecondary = lipgloss.Color("#979797")  
    TextMuted     = lipgloss.Color("#626262")  
  
    // Background colors  
    BgPrimary     = lipgloss.Color("#1A1A1A")  
    BgSecondary   = lipgloss.Color("#2A2A2A")  
    BgHighlight   = lipgloss.Color("#3A3A3A")  
)
```

Light Theme

```
var (  
    Primary    = lipgloss.Color("#5A00FF")  
    Secondary  = lipgloss.Color("#FF1493")  
    Accent     = lipgloss.Color("#0099CC")  
  
    Success    = lipgloss.Color("#00AA00")  
    Warning    = lipgloss.Color("#FF8800")  
    Error      = lipgloss.Color("#CC0000")  
    Info       = lipgloss.Color("#0066FF")  
  
    TextPrimary   = lipgloss.Color("#000000")  
    TextSecondary = lipgloss.Color("#666666")  
    TextMuted     = lipgloss.Color("#999999")  
  
    BgPrimary     = lipgloss.Color("#FFFFFF")  
    BgSecondary   = lipgloss.Color("#F5F5F5")  
    BgHighlight   = lipgloss.Color("#E5E5E5")  
)
```

Typography

Text Styles

```

// Headers
h1Style := lipgloss.NewStyle().
    Bold(true).
    Foreground(Primary).
    MarginTop(1).
    MarginBottom(1)

h2Style := lipgloss.NewStyle().
    Bold(true).
    Foreground(Secondary).
    MarginBottom(1)

// Body text
bodyStyle := lipgloss.NewStyle().
    Foreground(TextPrimary)

// Muted/secondary text
mutedStyle := lipgloss.NewStyle().
    Foreground(TextMuted).
    Italic(true)

// Code/monospace
codeStyle := lipgloss.NewStyle().
    Foreground(Accent).
    Background(BgSecondary).
    Padding(0, 1)

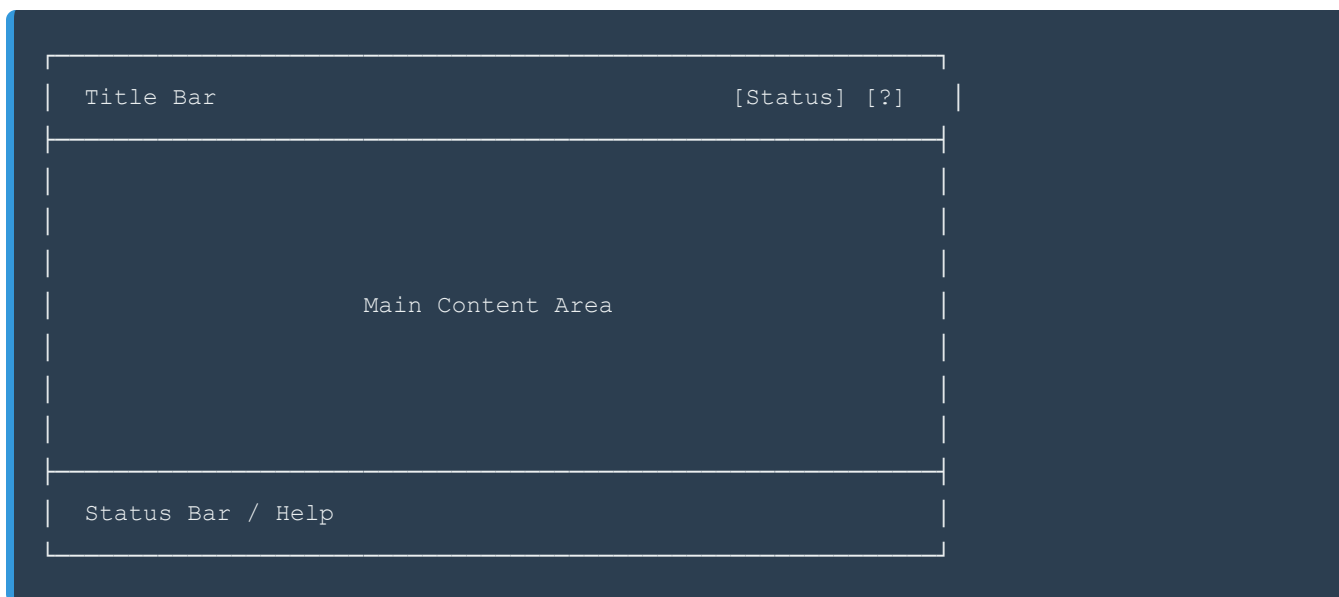
// Success/error messages
successStyle := lipgloss.NewStyle().
    Foreground(Success).
    Bold(true)

errorStyle := lipgloss.NewStyle().
    Foreground(Error).
    Bold(true)

```

Layout Patterns

1. Full-Screen Application



2. Sidebar + Content



3. Three-Column Layout

Title		
Left Panel	Center Panel	Right Panel
(List)	(Details)	(Properties)
Help		

4. Split Horizontal

Title
Top Panel
Bottom Panel

Status Indicators

```
// Loading
spinner := "" Loading..."

// Progress
progress := "██████████ 40%"

// Status icons
const (
    IconSuccess   = "✓"
    IconError      = "✗"
    IconWarning    = "⚠"
    IconInfo       = "i"
    IconRunning    = "🔄"
    IconPending    = "o"
    IconCompleted  = "●"
    IconSkipped    = "-"
)

// Usage
status := fmt.Sprintf("%s Done", IconSuccess)
status := fmt.Sprintf("%s Running...", IconRunning)
status := fmt.Sprintf("%s Failed", IconError)
```

UX Best Practices

Keyboard Navigation

Standard Keys **Navigation:** - `↑/k` - Move up - `↓/j` - Move down - `←/h` - Move left - `→/l` - Move right - `PgUp/Ctrl+u` - Page up - `PgDn/Ctrl+d` - Page down - `Home/g` - Go to top - `End/G` - Go to bottom **Selection:** - `Enter` - Select/confirm - `Space` - Toggle selection (multi-select) - `Tab` - Next field/section - `Shift+Tab` - Previous field/section - `Esc` - Cancel/back **Actions:** - `?` - Show help - `q/Ctrl+c` - Quit - `/` - Search/filter - `n/N` - Next/previous search result - `r` - Refresh - `e` - Edit - `d` - Delete - `a` - Add new

Application: - `Ctrl+p` - Command palette - `Ctrl+,` - Settings - `Ctrl+z` - Undo - `Ctrl+y`
- Redo ##### Vim-Style Bindings Many TUI apps support both arrow keys and vim-style: -
`h/j/k/l` for navigation - `gg/G` for top/bottom - `w/b` for word navigation (in text fields) - `:q`
to quit - `:w` to save ##### Custom Keybindings

```
type keyMap struct {
    Up      key.Binding
    Down    key.Binding
    Help    key.Binding
    Quit    key.Binding
}

var keys = keyMap{
    Up: key.NewBinding(
        key.WithKeys("up", "k"),
        key.WithHelp("↑/k", "move up"),
    ),
    Down: key.NewBinding(
        key.WithKeys("down", "j"),
        key.WithHelp("↓/j", "move down"),
    ),
    Help: key.NewBinding(
        key.WithKeys("?"),
        key.WithHelp("?", "toggle help"),
    ),
    Quit: key.NewBinding(
        key.WithKeys("q", "ctrl+c"),
        key.WithHelp("q", "quit"),
    ),
}
```

Mouse Support

Best Practices: 1. **Mouse is Optional:** All functionality accessible via keyboard 2. **Click Actions:** Single click to select, double click to activate 3. **Scroll Wheel:** Scroll content in focused panel 4. **Drag:** Resize panels, reorder lists (when applicable) 5. **Hover:** Show tooltips or highlight (if terminal supports) **Implementation:**

```
case tea.MouseMsg:
    switch msg.Button {
    case tea.MouseButtonLeft:
        if msg.Action == tea.MouseActionPress {
            // Handle click at msg.X, msg.Y
        }
    case tea.MouseButtonWheelUp:
        // Scroll up
    case tea.MouseButtonWheelDown:
        // Scroll down
    }
```

Help System

Inline Help Always show critical shortcuts in footer:

Content
↑/↓ navigate • enter select • / search • ? help • q quit

Full Help Screen Toggle with `?` key:

Keyboard Shortcuts		[Press ? to close]
Navigation:		
↑/k	Move up	
↓/j	Move down	
PgUp/Ctrl+u	Page up	
PgDn/Ctrl+d	Page down	
Actions:		
Enter	Select item	
/	Search	
r	Refresh	
q/Ctrl+c	Quit	
Help:		
?	Toggle this help screen	

Contextual Help Show relevant help based on current screen:

```
func (m model) getHelp() string {
    switch m.currentScreen {
    case screenDSLEditor:
        return "Ctrl+s save • Ctrl+r run • Esc back"
    case screenAgentPicker:
        return "Enter select • / filter • Esc back"
    default:
        return "? help • q quit"
    }
}
```

Error Handling & Validation

Real-Time Validation

```
Agent Name: my agent_  
✗ Name must contain only lowercase letters and hyphens
```

Error Messages **Good:**

```
✗ Error: Failed to parse DSL  
Line 5: Unexpected token 'then'  
Expected: operator (|, >>, &, ?)
```

Bad:

```
Error: Syntax error at line 5
```

Success Feedback

```
✓ Agent created successfully: my-agent  
Location: ~/.hekat/agents/my-agent.py  
Next: Run with 'hekat run my-agent'
```

Progressive Disclosure

Show Details on Demand **Collapsed:**

```
> Agent: my-agent [Expand ▼]
```

Expanded:

```
▼ Agent: my-agent [Collapse ▲]  
Description: Custom agent for data processing  
Created: 2024-10-20  
Runtime: Python 3.11  
Commands: process, analyze, report
```

Wizard-Style Forms Show one step at a time:

Step 1 of 3: Basic Information

Agent Name: _____

Description: _____

[Next >] [Cancel]

Accessibility

Screen Reader Support 1. **Avoid Fancy Unicode:** Use ASCII for critical info 2. **Semantic Structure:** Headers, lists, clearly marked 3. **Alt Text:** Describe visual elements 4. **Focus Indicators:** Clear focus state #### Color Blindness 1. **Don't Rely on Color Alone:** Use icons + text 2. **High Contrast:** Ensure readability 3. **Color-Blind Themes:** Provide alternatives
Example - Good:

```
✓ Success: Build completed
✗ Error: Tests failed
△ Warning: Deprecated API used
```

Example - Bad (color only):

```
Success: Build completed (green text only)
Error: Tests failed (red text only)
```

High-Contrast Mode

```
// Auto-detect terminal capabilities
if isMonochrome() {
    // Use bold/underline instead of color
    successStyle = lipgloss.NewStyle().Bold(true)
    errorStyle = lipgloss.NewStyle().Underline(true)
} else {
    successStyle = lipgloss.NewStyle().Foreground(green)
    errorStyle = lipgloss.NewStyle().Foreground(red)
}
```

Performance

Lazy Rendering Only render visible content:

```
// Bad: Render all 10,000 items
for _, item := range allItems {
    view += renderItem(item)
}

// Good: Render visible window
visibleStart := m.scrollTopOffset
visibleEnd := min(m.scrollTopOffset + m.height, len(allItems))
for i := visibleStart; i < visibleEnd; i++ {
    view += renderItem(allItems[i])
}
```

Debouncing Don't re-render on every keystroke:

```
case tea.KeyMsg:
    m.input.Update(msg)
    // Debounce search
    return m, tea.Tick(300*time.Millisecond, func(t time.Time) tea.Msg {
        return searchMsg{m.input.Value()}
    })
```

Virtual Scrolling For large lists, render only visible items:

```
type viewport struct {
    content      []string // All content
    height       int      // Visible height
    scrollTopOffset int     // Current scroll position
}

func (v viewport) View() string {
    start := v.scrollTopOffset
    end := min(start + v.height, len(v.content))
    return strings.Join(v.content[start:end], "\n")
}
```

Framework Feature Matrix

Textual - Deep Dive

Component Model:

```
from textual.app import App, ComposeResult
from textual.widgets import Button, Input, DataTable

class MyApp(App):
    CSS_PATH = "app.tcss" # External stylesheet

    def compose(self) -> ComposeResult:
        """Build UI"""
        yield Input(placeholder="Enter text...")
        yield Button("Submit", id="submit")
        yield DataTable()

    def on_button_pressed(self, event: Button.Pressed) -> None:
        """Handle button click"""
        if event.button.id == "submit":
            self.handle_submit()
```

CSS Styling (TCSS):

```
/* app.tcss */
Screen {
    background: $surface;
}

Input {
    dock: top;
    margin: 1;
    border: heavy $primary;
}

Button {
    margin: 1;
    width: 100%;
}

Button: hover {
    background: $accent;
}

DataTable {
    height: 100%;
    border: solid $primary;
}
```

Key Widgets Input & TextArea:

```

from textual.widgets import Input, TextArea

# Single-line input
input = Input(
    placeholder="Agent name...",
    max_length=64,
    valid_empty=False,
)

# Multi-line editor
editor = TextArea(
    language="python", # Syntax highlighting
    theme="monokai",
    show_line_numbers=True,
)

```

DataTable:

```

from textual.widgets import DataTable

table = DataTable()
table.add_columns("Agent", "Status", "Duration")
table.add_row("agent-1", "✓ Done", "2.3s")
table.add_row("agent-2", "⚙ Running", "1.1s")

# Cursor support
table.cursor_type = "row"
table.focus()

```

Tree:

```

from textual.widgets import Tree

tree = Tree("Root")
node = tree.root.add("Child 1")
node.add_leaf("Grandchild 1")
node.add_leaf("Grandchild 2")
tree.root.add("Child 2")

```

TabbedContent:

```

from textual.widgets import TabbedContent, TabPane

with TabbedContent():
    with TabPane("DSL", id="dsl"):
        yield TextArea()
    with TabPane("Agents", id="agents"):
        yield DataTable()
    with TabPane("Logs", id="logs"):
        yield RichLog()

```

Layout System **Docking**:

```

Header {
    dock: top;
    height: 3;
}

Footer {
    dock: bottom;
    height: 1;
}

Sidebar {
    dock: left;
    width: 30;
}

```

Grid:

```

Container {
    layout: grid;
    grid-size: 3; /* 3 columns */
    grid-rows: 1fr 2fr 1fr;
    grid-columns: 1fr 2fr 1fr;
}

```

Horizontal/Vertical:

```
Container {  
    layout: horizontal; /* or vertical */  
    height: auto;  
}
```

Event Handling

```
from textual.app import App  
from textual.events import Key, Click  
  
class MyApp(App):  
    def on_key(self, event: Key) -> None:  
        """Global key handler"""  
        if event.key == "ctrl+s":  
            self.save()  
        elif event.key == "f1":  
            self.show_help()  
  
    def on_click(self, event: Click) -> None:  
        """Global click handler"""  
        self.log(f"Clicked at {event.x}, {event.y}")  
  
    def on_button_pressed(self, event: Button.Pressed) -> None:  
        """Specific widget event"""  
        button_id = event.button.id  
        self.log(f"Button pressed: {button_id}")
```

Commands & Actions

```
from textual.app import App
from textual.command import Provider

class MyApp(App):
    COMMANDS = {MyCommandProvider}

    def action_save(self) -> None:
        """Actions can be called from anywhere"""
        self.save()

    def action_quit(self) -> None:
        self.exit()

# Command palette (Ctrl+P)
class MyCommandProvider(Provider):
    async def search(self, query: str) -> Hits:
        if query.startswith("save"):
            yield Hit("save", "Save current file", self.app.action_save)
        if query.startswith("quit"):
            yield Hit("quit", "Quit application", self.app.action_quit)
```

Rich - Deep Dive

Console

```

from rich.console import Console

console = Console()

# Styled output
console.print("[bold magenta]Hekat[/] - Agent Orchestration DSL")
console.print("✓ Build completed", style="green")

# Tables
from rich.table import Table

table = Table(title="Agents")
table.add_column("Name", style="cyan")
table.add_column("Status", style="magenta")
table.add_column("Duration", justify="right", style="green")

table.add_row("agent-1", "✓ Done", "2.3s")
table.add_row("agent-2", "⚙ Running", "1.1s")
console.print(table)

```

Progress

```

from rich.progress import Progress

with Progress() as progress:
    task1 = progress.add_task("[red]Downloading...", total=100)
    task2 = progress.add_task("[green]Processing...", total=100)

    while not progress.finished:
        progress.update(task1, advance=0.5)
        progress.update(task2, advance=0.3)
        time.sleep(0.02)

```

Live Display

```
from rich.live import Live
from rich.table import Table

def generate_table():
    table = Table()
    table.add_column("Status")
    table.add_column("Progress")
    # ... populate table
    return table

with Live(generate_table(), refresh_per_second=4) as live:
    while True:
        time.sleep(0.25)
        live.update(generate_table()) # Update display
```

Syntax Highlighting

```
from rich.syntax import Syntax

code = """
def hello():
    print("Hello, world!")
"""

syntax = Syntax(code, "python", theme="monokai", line_numbers=True)
console.print(syntax)
```

Markdown

```
from rich.markdown import Markdown

markdown_text = """
# Hekat DSL

## Features
- Agent orchestration
- Parallel execution
- Streaming support
"""

md = Markdown(markdown_text)
console.print(md)
```

Prompt Toolkit - Deep Dive

Basic Input

```
from prompt_toolkit import prompt
from prompt_toolkit.completion import WordCompleter

# Simple input
name = prompt("Enter name: ")

# With completion
agent_completer = WordCompleter(['agent-1', 'agent-2', 'agent-3'])
agent = prompt("Select agent: ", completer=agent_completer)

# Password
password = prompt("Password: ", is_password=True)
```

Application

```
from prompt_toolkit.application import Application
from prompt_toolkit.key_binding import KeyBindings
from prompt_toolkit.layout import Layout
from prompt_toolkit.widgets import TextArea

# Key bindings
kb = KeyBindings()

@kb.add('c-c')
@kb.add('c-q')
def _(event):
    event.app.exit()

# Text area
text_area = TextArea(text="Initial text")

# Layout
layout = Layout(text_area)

# Application
app = Application(
    layout=layout,
    key_bindings=kb,
    full_screen=True,
)

app.run()
```

Forms

```

from prompt_toolkit.shortcuts import radiolist_dialog, checkboxes_dialog

# Radio buttons
runtime = radiolist_dialog(
    title="Select Runtime",
    text="Choose a runtime for your agent:",
    values=[
        ("python", "Python 3.11"),
        ("node", "Node.js 20"),
        ("go", "Go 1.21"),
    ],
).run()

# Checkboxes
features = checkboxes_dialog(
    title="Select Features",
    text="Enable features:",
    values=[
        ("async", "Async support"),
        ("parallel", "Parallel execution"),
        ("streaming", "Streaming output"),
    ],
).run()

```

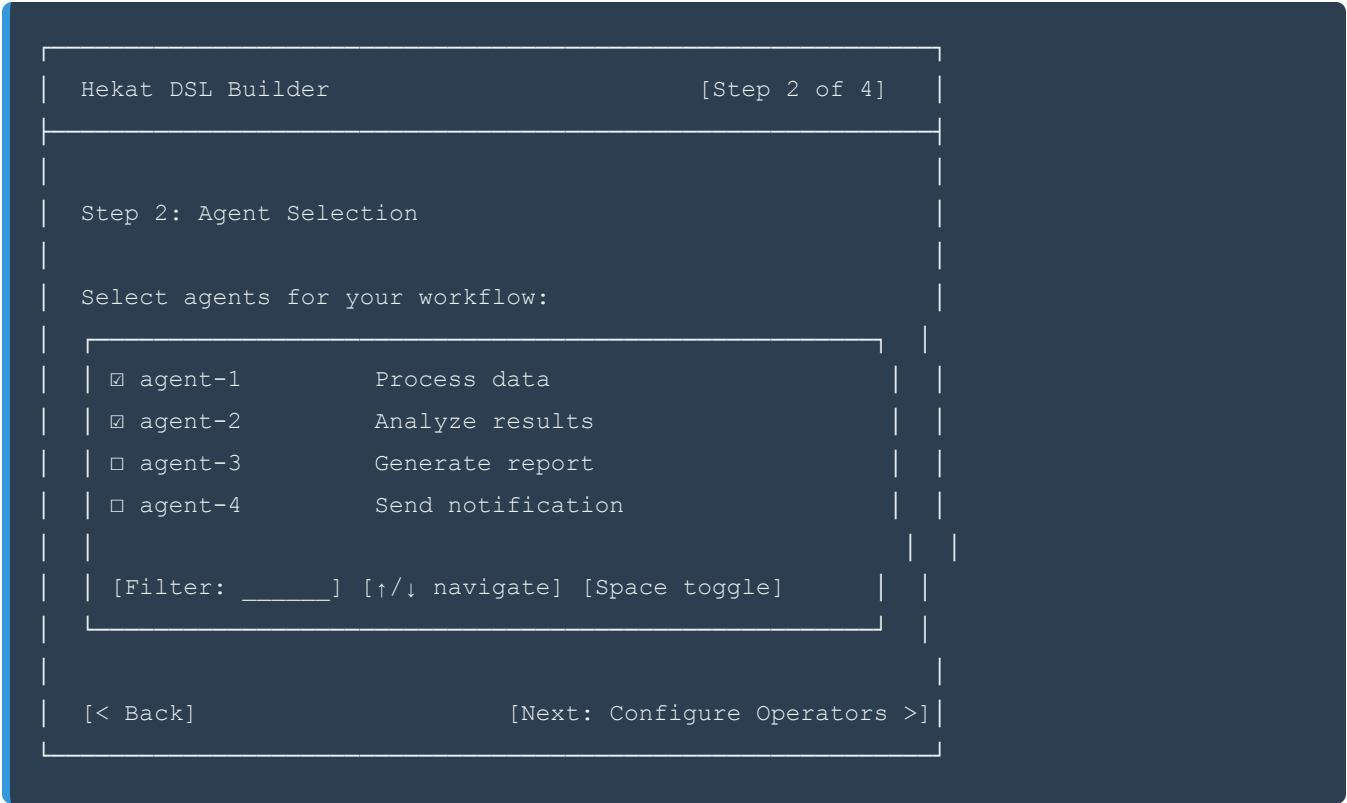
Framework Selection for Hekat

Recommendation: Textual **Why Textual?** 1. **Complete TUI Framework:** Everything needed in one package 2. **Python Native:** Direct integration with Hekat codebase 3. **Modern API:** Web-inspired, familiar to developers 4. **Rich Widgets:** DataTable, TextArea, Tree, etc. cover all needs 5. **CSS Styling:** Maintainable, familiar styling approach 6. **Async First:** Matches Hekat's async architecture 7. **Active Development:** Frequent updates, strong community 8. **Browser Support:** Can export TUI to web (future bonus) 9. **Excellent Docs:** Comprehensive tutorials and examples **Use Rich for Enhancement:** - Syntax highlighting in TextArea - Beautiful console output for CLI commands - Markdown rendering for help/docs - Progress bars for long operations **Don't Use Prompt Toolkit:** - More manual work required - Better suited for CLI prompts than full TUI - Less comprehensive widget library ---

Hekat TUI Vision & Features

Interactive DSL Builder (Phase 5)

Feature: Wizard-Style Workflow Creator **Purpose:** Guide users through creating Hekat DSL workflows without writing code **Layout:**



Steps: **Step 1: Workflow Metadata** - Name - Description - Tags - Timeout **Step 2: Agent Selection** - Multi-select from available agents - Search/filter agents - Show agent descriptions - Preview agent commands **Step 3: Operator Configuration** - Choose operators (|, >>, &, ?) - Set streaming vs batch - Configure parallelism - Set timeouts **Step 4: Review & Generate** - Preview generated DSL - Syntax highlighting - Validate DSL - Save to file **Implementation:**

```

from textual.app import App, ComposeResult
from textual.widgets import Header, Footer, Button, Static
from textual.containers import Container, VerticalScroll
from textual.screen import Screen

class Step1_Metadata(Screen):
    def compose(self) -> ComposeResult:
        yield Header()
        yield Container(
            Static("Step 1: Workflow Metadata"),
            Input(placeholder="Workflow name", id="name"),
            TextArea(placeholder="Description", id="desc"),
            Input(placeholder="Tags (comma-separated)", id="tags"),
            Input(placeholder="Timeout (seconds)", id="timeout"),
            Button("Next >", variant="primary", id="next"),
        )
        yield Footer()

class Step2_AgentSelection(Screen):
    def compose(self) -> ComposeResult:
        yield Header()
        yield Container(
            Static("Step 2: Agent Selection"),
            Input(placeholder="Filter agents...", id="filter"),
            DataTable(id="agents"), # Multi-select enabled
            Button("< Back", id="back"),
            Button("Next >", variant="primary", id="next"),
        )
        yield Footer()

class DSLBuilderApp(App):
    def on_mount(self) -> None:
        self.push_screen(Step1_Metadata())

    def on_button_pressed(self, event: Button.Pressed) -> None:
        if event.button.id == "next":
            self.advance_step()
        elif event.button.id == "back":
            self.previous_step()

```

Feature: Autocomplete DSL Editor **Purpose:** Write DSL with real-time autocomplete and validation **Layout:**

```
DSL Editor - my-workflow.hekat [Ctrl+S Save]

1 workflow data-pipeline:
2     agent-1 | agent-2 >> agent-3
3
4     # Suggestion: Use & for parallel
5     agent-
6
7     agent-1
8     agent-2
9     agent-3
10    agent-4
11
12    ...

✓ Valid DSL • 2 agents • 3 operators [Ctrl+R Run]
```

Features: 1. **Syntax Highlighting:** Color-code keywords, operators, agents 2. **Autocomplete:** Suggest agents, operators, keywords 3. **Real-Time Validation:** Show errors immediately 4. **Operator Hints:** Suggest appropriate operators 5. **Line Numbers:** Easy navigation 6. **Status Bar:** Show parse status, token count **Implementation:**

```

from textual.widgets import TextArea
from textual.validation import Function, ValidationResult

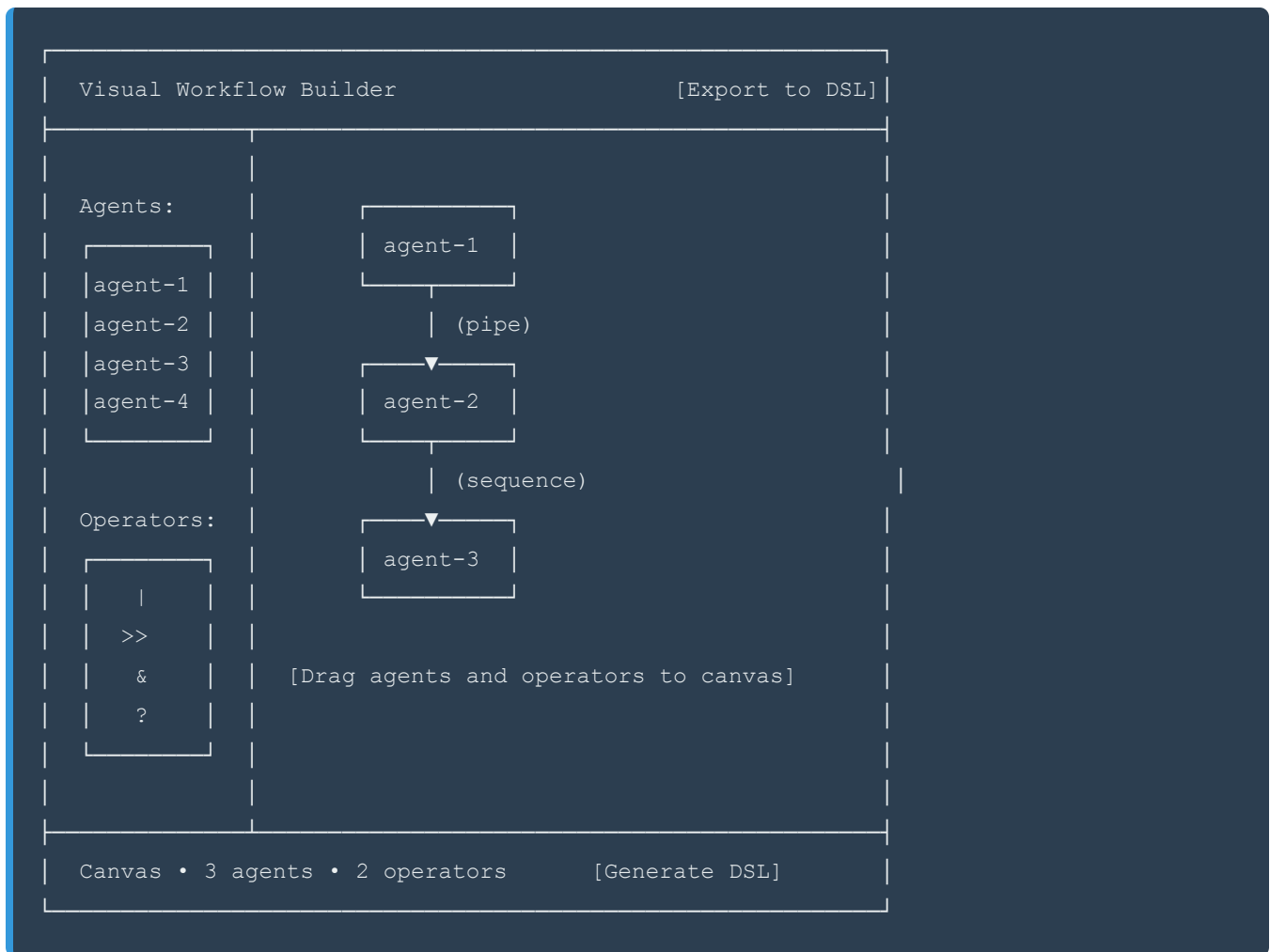
class DSLEditor(TextArea):
    def __init__(self):
        super().__init__(
            language="python", # Closest syntax highlighting
            theme="monokai",
            show_line_numbers=True,
        )

    def on_text_area_changed(self) -> None:
        """Validate on change"""
        dsl_text = self.text
        errors = self.validate_dsl(dsl_text)
        if errors:
            self.show_errors(errors)
        else:
            self.show_valid()

    def get_autocomplete_suggestions(self, cursor_pos):
        """Return suggestions based on cursor position"""
        # Parse DSL, determine context
        # Return relevant suggestions (agents, operators, keywords)
        pass

```

Feature: Visual Workflow Builder **Purpose:** Drag-and-drop workflow construction **Layout:**



Note: This is ambitious and may be Phase 7+

Syntax Highlighting & Validation (Phase 5)

Real-Time DSL Preview **Purpose:** See DSL with syntax highlighting while typing

Implementation:

```

from rich.syntax import Syntax
from textual.widgets import Static

class DSLPreview(Static):
    def update_dsl(self, dsl_text: str):
        """Render DSL with syntax highlighting"""
        syntax = Syntax(
            dsl_text,
            "python", # Use Python syntax as closest match
            theme="monokai",
            line_numbers=True,
        )
        self.update(syntax)

```

Error Highlighting **Purpose:** Show errors inline with squiggly underlines

```

1 workflow my-flow:
2     agent-1 | agent-2
3         ^^^^^^ Unknown agent "agent-2"
4             Did you mean "agent-2"?
5
6     agent-1 >> agent-2 |
7         ^ Incomplete operator chain

```

Validation Status **Bottom Status Bar:**

```
✓ Valid DSL • 3 agents • 4 operators • Est. tokens: 1,234
```

With Errors:

```
✗ 2 errors • Line 2: Unknown agent • Line 6: Incomplete chain
```

Execution Dashboard (Phase 6)

Real-Time Progress Tracking **Purpose:** Monitor workflow execution with live updates

Layout:


```

from textual.widgets import ProgressBar, DataTable, RichLog

class ExecutionDashboard(Screen):
    def compose(self) -> ComposeResult:
        yield Header()
        yield ProgressBar(total=100, id="overall")
        yield DataTable(id="agents")
        yield RichLog(id="logs", auto_scroll=True)
        yield Footer()

    async def watch_execution(self, workflow):
        """Update dashboard in real-time"""
        while workflow.running:
            # Update progress
            progress = workflow.get_progress()
            self.query_one("#overall", ProgressBar).update(progress)

            # Update agent table
            table = self.query_one("#agents", DataTable)
            for agent in workflow.agents:
                table.update_row(agent.id, [
                    agent.name,
                    agent.status,
                    agent.duration,
                    agent.token_count,
                ])

            # Stream logs
            logs = self.query_one("#logs", RichLog)
            for log_entry in workflow.get_new_logs():
                logs.write(log_entry)

            await asyncio.sleep(0.1)  # 100ms refresh

```

Parallel Stream Visualization **Purpose:** Show parallel agent execution **Layout:**


```
Debugger [Step] [Continue] [Stop]

Workflow: data-pipeline

1 workflow data-pipeline:
2 ► agent-1 | agent-2 >> agent-3
3 # Currently at agent-2

Current State:

agent-1 output:
{"data": [1, 2, 3], "count": 3}

agent-2 input:
Stream([1, 2, 3])

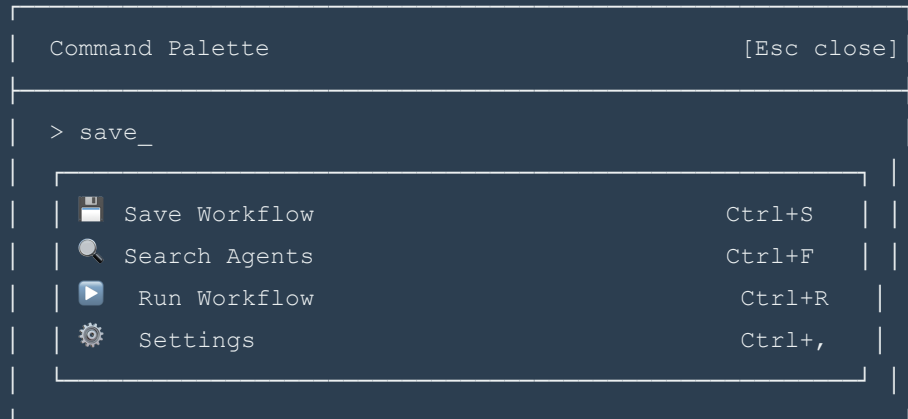
Variables:
count = 3
tokens_used = 1234

[F5] Continue • [F10] Step Over • [F11] Step Into
```

Features: 1. **Breakpoints:** Set breakpoints at specific agents 2. **Step Through:** Execute one agent at a time 3. **Inspect State:** View input/output, variables 4. **Continue:** Resume normal execution 5. **Stack Trace:** Show execution history

Help System (Phase 5)

Command Palette **Trigger:** `Ctrl+P` or `/`



Implementation:

```
from textual.command import Provider, Hit

class HekatCommandProvider(Provider):
    async def search(self, query: str) -> Hits:
        """Fuzzy search commands"""
        matcher = self.matcher(query)

        commands = [
            ("save", "Save Workflow", self.app.action_save, "Ctrl+S"),
            ("run", "Run Workflow", self.app.action_run, "Ctrl+R"),
            ("debug", "Debug Workflow", self.app.action_debug, "Ctrl+D"),
            ("help", "Show Help", self.app.action_help, "F1"),
        ]

        for cmd_id, cmd_name, cmd_action, shortcut in commands:
            score = matcher.match(cmd_name)
            if score > 0:
                yield Hit(
                    score,
                    matcher.highlight(cmd_name),
                    cmd_action,
                    help=shortcut,
                )
```

Documentation Browser **Purpose:** Browse Hekat documentation without leaving TUI

```
python | | | agent-1 | agent-2 # Stream data | | >> |
```

```
python | | agent-1 >> agent-2 # Full output | |
```

Implementation with Glamour/Rich:

```

from rich.markdown import Markdown
from textual.widgets import Tree, Static

class DocsBrowser(Screen):
    def compose(self) -> ComposeResult:
        with Container():
            yield Tree("Documentation", id="topics")
            yield Static(id="content")

    def on_tree_node_selected(self, event) -> None:
        """Load and render markdown"""
        topic = event.node.data
        markdown_content = load_doc(topic)

        # Render with Rich
        md = Markdown(markdown_content)
        self.query_one("#content", Static).update(md)

```

Implementation Roadmap

Phase 5: Interactive DSL Builder (Weeks 1-3)

Week 1: Foundation - Set up Textual app structure - Implement basic navigation - Create app skeleton with Header/Footer - Design color scheme and styles - Create TCSS stylesheet

Week 2: DSL Builder Wizard - Step 1: Metadata form (Huh-style) - Step 2: Agent multi-select with search - Step 3: Operator configuration - Step 4: Review and generate DSL - Navigation between steps - Data persistence between steps

Week 3: DSL Editor - TextArea with syntax highlighting - Real-time validation - Autocomplete suggestions - Line numbers and status bar - Save/load functionality - Integration with existing DSL parser

Deliverables: - `hekat tui` command launches TUI - Interactive wizard creates valid DSL - Code editor with validation - Help system with keyboard shortcuts

Phase 6: Execution Dashboard (Weeks 4-6)

Week 4: Dashboard Layout - Progress bar component - Agent status table (DataTable) - Live log viewer (RichLog) - Control buttons (pause/stop/restart) - Real-time updates (async) **Week 5: Execution Integration** - Connect to workflow execution engine - Stream agent logs to TUI - Update status in real-time - Token counting and display - Time tracking (elapsed/remaining) **Week 6: Advanced Features** - Parallel execution visualization - Multiple workflow tabs (TabbedContent) - Execution history browser - Export logs to file - Error handling and display **Deliverables:** - `hekat run workflow.hekat --tui` launches dashboard - Real-time execution monitoring - Log streaming and filtering - Pause/resume/stop controls

Phase 7: Debugging & Advanced (Weeks 7-9)

Week 7: Debugger - Step-through execution - Breakpoint support - State inspection - Variable viewer - Stack trace display **Week 8: Enhanced Help** - Command palette (Ctrl+P) - Documentation browser - Agent library browser - Examples explorer - Interactive tutorials **Week 9: Polish & Testing** - Performance optimization - Keyboard shortcut refinement - Accessibility improvements - Comprehensive testing - Documentation **Deliverables:** - `hekat debug workflow.hekat` launches debugger - Command palette with all actions - Integrated documentation browser - Production-ready, tested TUI ---

Code Examples & Patterns

Complete Textual App Example

```
# hekat/tui/app.py
from textual.app import App, ComposeResult
from textual.widgets import Header, Footer, Button, Input, DataTable
from textual.containers import Container, Horizontal, Vertical
from textual.screen import Screen
from textual import on

# CSS
CSS = """
Screen {
    background: $surface;
}

Header {
    dock: top;
    height: 3;
    background: $primary;
}

Footer {
    dock: bottom;
    height: 1;
    background: $panel;
}

Container {
    height: 100%;
    padding: 1;
}

Input {
    margin: 1 0;
}

Button {
    margin: 1;
}
```

```

DataTable {
    height: 1fr;
    border: solid $primary;
}
"""

class AgentSelector(Screen):
    """Agent selection screen"""

    CSS = CSS

    def compose(self) -> ComposeResult:
        yield Header()
        yield Container(
            Input(placeholder="Filter agents...", id="filter"),
            DataTable(id="agents"),
            Horizontal(
                Button("Cancel", variant="error", id="cancel"),
                Button("Select", variant="success", id="select"),
            ),
        )
        yield Footer()

    def on_mount(self) -> None:
        """Initialize table"""
        table = self.query_one("#agents", DataTable)
        table.add_columns("Agent", "Description", "Runtime")
        table.add_rows([
            ("agent-1", "Data processor", "Python"),
            ("agent-2", "Analyzer", "Python"),
            ("agent-3", "Reporter", "Go"),
        ])
        table.cursor_type = "row"
        table.focus()

    @on(Input.Changed, "#filter")
    def filter_agents(self, event: Input.Changed) -> None:
        """Filter agents as user types"""
        query = event.value.lower()
        table = self.query_one("#agents", DataTable)
        # Filter logic here

```

```

@on(Button.Pressed, "#select")
def select_agents(self) -> None:
    """Handle selection"""
    table = self.query_one("#agents", DataTable)
    selected_row = table.cursor_row
    # Process selection
    self.dismiss(selected_row)

@on(Button.Pressed, "#cancel")
def cancel(self) -> None:
    self.dismiss(None)

class HekatTUI(App):
    """Main Hekat TUI application"""

    CSS = CSS
    TITLE = "Hekat DSL Builder"
    BINDINGS = [
        ("q", "quit", "Quit"),
        ("?", "help", "Help"),
        ("ctrl+s", "save", "Save"),
        ("ctrl+r", "run", "Run"),
    ]

    def compose(self) -> ComposeResult:
        yield Header()
        yield Container(
            # Main content
        )
        yield Footer()

    def action_quit(self) -> None:
        """Quit application"""
        self.exit()

    def action_help(self) -> None:
        """Show help"""
        # Show help screen
        pass

    def action_save(self) -> None:
        """Save workflow"""
        # Save logic

```

```
        pass

    def action_run(self) -> None:
        """Run workflow"""
        # Launch execution dashboard
        pass

if __name__ == "__main__":
    app = HekatTUI()
    app.run()
```

DSL Editor with Validation

```
from textual.widgets import TextArea
from textual.validation import Function, ValidationResult
from hekat.parser import parse_dsl, DSLError

class DSLEditor(TextArea):
    """DSL editor with real-time validation"""

    def __init__(self):
        super().__init__(
            language="python",
            theme="monokai",
            show_line_numbers=True,
        )
        self.errors = []

    def watch_text(self, new_text: str) -> None:
        """Validate DSL on change"""
        try:
            ast = parse_dsl(new_text)
            self.errors = []
            self.styles.border = ("solid", "green")
        except DSLError as e:
            self.errors = [e]
            self.styles.border = ("solid", "red")

    def get_status(self) -> str:
        """Return validation status"""
        if self.errors:
            error = self.errors[0]
            return f"✗ Error line {error.line}: {error.message}"
        else:
            return "✓ Valid DSL"
```

Live Execution Dashboard

```
from textual.widgets import ProgressBar, DataTable, RichLog
from textual.screen import Screen
import asyncio

class ExecutionDashboard(Screen):
    """Real-time workflow execution dashboard"""

    def compose(self) -> ComposeResult:
        yield Header()
        yield ProgressBar(total=100, show_percentage=True, id="progress")
        yield DataTable(id="agents")
        yield RichLog(id="logs", auto_scroll=True)
        yield Footer()

    def on_mount(self) -> None:
        """Initialize dashboard"""
        # Setup agent table
        table = self.query_one("#agents", DataTable)
        table.add_columns("Agent", "Status", "Duration", "Tokens")

        # Start monitoring
        self.set_interval(0.1, self.update_dashboard)

    def update_dashboard(self) -> None:
        """Update dashboard (called every 100ms)"""
        # Update progress bar
        progress = self.workflow.get_progress()
        self.query_one("#progress", ProgressBar).update(progress=progress)

        # Update agent table
        table = self.query_one("#agents", DataTable)
        for agent in self.workflow.agents:
            row_key = agent.id
            if not table.is_valid_row_key(row_key):
                table.add_row(
                    agent.name,
                    self.status_icon(agent.status),
                    f"{agent.duration:.1f}s",
                    str(agent.tokens),
                    key=row_key,
```

```

    )
    else:
        table.update_cell(
            row_key,
            "Status",
            self.status_icon(agent.status),
        )
        table.update_cell(
            row_key,
            "Duration",
            f"{agent.duration:.1f}s",
        )
        table.update_cell(
            row_key,
            "Tokens",
            str(agent.tokens),
        )

    # Stream logs
    logs = self.query_one("#logs", RichLog)
    for log_entry in self.workflow.get_new_logs():
        logs.write(log_entry)

    @staticmethod
    def status_icon(status: str) -> str:
        """Get status icon"""
        return {
            "pending": "o",
            "running": "e",
            "done": "✓",
            "error": "✗",
        }.get(status, "?")

```

Command Palette

```
from textual.command import Provider, Hit
from textual.types import IgnoreReturnCallbackType

class HekatCommands(Provider):
    """Command palette for Hekat"""

    async def search(self, query: str) -> Hits:
        """Search commands"""
        matcher = self.matcher(query)

        commands = [
            ("new", "New Workflow", self.app.action_new, "Ctrl+N"),
            ("open", "Open Workflow", self.app.action_open, "Ctrl+O"),
            ("save", "Save Workflow", self.app.action_save, "Ctrl+S"),
            ("run", "Run Workflow", self.app.action_run, "Ctrl+R"),
            ("debug", "Debug Workflow", self.app.action_debug, "Ctrl+D"),
            ("agents", "Browse Agents", self.app.action_agents, "Ctrl+A"),
            ("help", "Show Help", self.app.action_help, "F1"),
            ("settings", "Settings", self.app.action_settings, "Ctrl+,",),
        ]

        for cmd_id, name, action, shortcut in commands:
            score = matcher.match(name)
            if score > 0:
                yield Hit(
                    score,
                    matcher.highlight(name),
                    action,
                    help=shortcut,
                )

class HekatTUI(App):
    COMMANDS = {HekatCommands} # Register provider

    # Enable command palette with Ctrl+P
    BINDINGS = [
        ("ctrl+p", "command_palette", "Commands"),
    ]
```

Resources & References

Charm Ecosystem

Official: - Bubble Tea: <https://github.com/charmbracelet/bubbletea> - Bubbles: <https://github.com/charmbracelet/bubbles> - Lip Gloss: <https://github.com/charmbracelet/lipgloss> - Glamour: <https://github.com/charmbracelet/glamour> - Huh: <https://github.com/charmbracelet/huh> - Charm.sh: <https://charm.sh> **Tutorials:** - Bubble Tea Tutorial: <https://github.com/charmbracelet/bubbletea/tree/master/tutorials> - Building TUIs with Go: <https://leg100.github.io/en/posts/building-bubbletea-programs/> - Tips for Bubble Tea: <https://themarkokovacevic.com/posts/terminal-ui-with-bubbletea/> **Videos:** - Charm.sh YouTube: <https://www.youtube.com/@charmcli> - Making CLIs Glamorous (Go Time podcast): <https://changelog.com/gotime/222> **Applications:** - Glow: <https://github.com/charmbracelet/glow> - VHS: <https://github.com/charmbracelet/vhs> - Soft Serve: <https://github.com/charmbracelet/soft-serve> - GitHub CLI: <https://github.com/cli/cli>

Textual (Python)

Official: - Textual: <https://github.com/Textualize/textual> - Documentation: <https://textual.textualize.io/> - Tutorial: <https://textual.textualize.io/tutorial/> - Widget Gallery: https://textual.textualize.io/widget_gallery/ **Learning Resources:** - Real Python Tutorial: <https://realpython.com/python-textual/> - Textual Definitive Guide: <https://dev.to/wiseai/textual-the-definitive-guide-part-1-1i0p> - ArjanCodes Guide: <https://arjancodes.com/blog/textual-python-library/> - Mathspp TODO Tutorial: <https://mathspp.com/blog/textual-tutorial-build-a-todo-app-in-python> **Blog Posts:** - 7 Things I Learned: <https://www.textualize.io/blog/7-things-ive-learned-building-a-modern-tui-framework/> - Algorithms for Performance: <https://textual.textualize.io/blog/2024/12/12/algorithms-for-high-performance-terminal-apps/> - Anatomy of TUI: <https://textual.textualize.io/blog/2024/09/15/anatomy-of-a-textual-user-interface/> **Applications:** - Posting (API client): <https://github.com/darrenburns/posting> -

Logmerger: <https://github.com/ptmcg/logmerger> - Textual TODO:
<https://github.com/Textualize/textual-todo>

Rich (Python)

Official: - Rich: <https://github.com/Textualize/rich> - Documentation: <https://rich.readthedocs.io/>

Tutorials: - Rich Library Guide: <https://medium.com/@ahmedharabi/get-started-with-the-python-rich-library-2736b1b57941> - 12 Ways to Beautify:

<https://medium.com/@jainsnehasj6/a-practical-guide-to-rich-12-ways-to-instantly-beautify-your-python-terminal-3a4a3434d04a> - FreeCodeCamp:

<https://www.freecodecamp.org/news/use-the-rich-library-in-python/>

Prompt Toolkit (Python)

Official: - Prompt Toolkit: <https://github.com/prompt-toolkit/python-prompt-toolkit> -

Documentation: <https://python-prompt-toolkit.readthedocs.io/> **Tutorials:** - Building Rich CLIs:

<https://www.w3computing.com/articles/python-clis-rich-user-interfaces-prompt-toolkit/>

TUI Design Resources

Best Practices: - Awesome TUIs: <https://github.com/rothgar/awesome-tuis> - TUI Guidelines:

<https://ux.stackexchange.com/questions/83400/guidelines-for-console-ui-cli> - Keyboard

Navigation: <https://www.chhs.colostate.edu/accessibility/best-practices-how-tos/keyboard-navigation/> **Inspiration:** - Terminal Trove: <https://terminaltrove.com/> - Text Mode Applications:

https://en.wikipedia.org/wiki/Text-based_user_interface ---

Appendix: Quick Reference

Textual Cheat Sheet

```
# App structure
from textual.app import App, ComposeResult
from textual.widgets import Button, Input, DataTable
from textual.containers import Container

class MyApp(App):
    def compose(self) -> ComposeResult:
        yield Button("Click me")

    def on_button_pressed(self) -> None:
        self.notify("Clicked!")

# Widgets
Button("Label", variant="primary", id="my-btn")
Input(placeholder="Text...", max_length=50)
TextArea(language="python", theme="monokai")
DataTable()
RichLog(auto_scroll=True)
ProgressBar(total=100)
Tree("Root")

# Events
def on_button_pressed(self, event: Button.Pressed) -> None: ...
def on_input_changed(self, event: Input.Changed) -> None: ...
def on_key(self, event: Key) -> None: ...

# Reactive
from textual.reactive import reactive

class Widget(Static):
    count = reactive(0)

    def watch_count(self, new_value): ...

# CSS (TCSS)
"""
Button {
```

```
        background: $primary;
        color: $text;
        margin: 1;
        border: solid $accent;
    }

    Button: hover {
        background: $accent;
    }
    """

# Commands
BINDINGS = [
    ("q", "quit", "Quit"),
    ("ctrl+s", "save", "Save"),
]

def action_quit(self) -> None: ...
def action_save(self) -> None: ...

# Screens
self.push_screen(MyScreen())
self.pop_screen()
self.switch_screen(MyScreen())
```

Rich Cheat Sheet

```
from rich.console import Console
from rich.table import Table
from rich.progress import Progress
from rich.syntax import Syntax
from rich.markdown import Markdown

console = Console()

# Styled output
console.print("[bold red>Error![/] Something went wrong")

# Tables
table = Table(title="Data")
table.add_column("Name")
table.add_column("Value")
table.add_row("Item 1", "100")
console.print(table)

# Progress
with Progress() as progress:
    task = progress.add_task("Processing...", total=100)
    progress.update(task, advance=10)

# Syntax
syntax = Syntax(code, "python", theme="monokai", line_numbers=True)
console.print(syntax)

# Markdown
md = Markdown("# Title\n\nContent")
console.print(md)
```

Color Palette Reference

```
# Charm-inspired colors
PRIMARY    = "#7D56F4"  # Purple
SECONDARY  = "#FF6AC1"  # Pink
ACCENT     = "#00D9FF"  # Cyan
SUCCESS    = "#04B575"  # Green
WARNING    = "#FFAA00"  # Orange
ERROR      = "#EE4B2B"  # Red
INFO       = "#00A9FF"  # Blue

# Dark theme
BG_DARK    = "#1A1A1A"
BG_DARK_ALT = "#2A2A2A"
TEXT_DARK  = "#FFFFFF"
TEXT_DARK_DIM = "#979797"

# Light theme
BG_LIGHT    = "#FFFFFF"
BG_LIGHT_ALT = "#F5F5F5"
TEXT_LIGHT  = "#000000"
TEXT_LIGHT_DIM = "#666666"
```

Conclusion

This comprehensive research provides everything needed to build a world-class TUI for Hekat using Textual and Rich. The Charm ecosystem demonstrates best-in-class TUI design patterns that can be replicated in Python, and Textual provides all the necessary tools to achieve similar polish and functionality. **Next Steps:** 1. Prototype basic Textual app with DSL editor 2.

Implement wizard-style workflow builder 3. Build execution dashboard with real-time updates 4. Polish and test extensively 5. Deploy as `hekat tui` command The TUI will significantly enhance Hekat's usability, making agent orchestration accessible to both power users and newcomers through visual, interactive interfaces. --- **Document Status:** Complete ✓

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